LLM-Based Antipattern Detection in UML Use Case Models: A Synthetic Data Generation and Fine-Tuning Approach

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Abstract

Antipatterns in UML use case models degrade software design quality and impede communication between stakeholders, yet their detection has received comparatively little automated support. We present a complete pipeline for fine-tuning an LLM to detect the *Functional Decomposition via* «include» *relationship* antipattern in PlantUML use case diagrams. The pipeline uses Claude Opus, a frontier LLM, to synthesize 194 labeled diagram pairs (each consisting of an antipattern version and its correctly refactored counterpart) across a diverse set of application domains. Qwen2.5-Coder-3B-Instruct, a compact LLM, is then fine-tuned on this dataset using Low-Rank Adaptation (LoRA), achieving 91.0% detection accuracy and an instance-level recall of 0.87 on a held-out, domain-stratified test set. The paper details the full pipeline from synthetic data generation to fine-tuning and evaluation.

Keywords

UML, use case diagrams, antipattern detection, LLM, fine-tuning, synthetic data, PlantUML, LoRA

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1 Introduction

Unified Modeling Language (UML) use case diagrams are a primary artifact in requirements engineering. They describe the functional scope of a system in terms of actors, use cases, and the relationships between them («include», «extend», and generalisation). Despite their conceptual simplicity, practitioners regularly introduce structural antipatterns that reduce diagram clarity, misrepresent system behavior, or create maintenance obstacles [3, 14].

One particularly common antipattern is *Functional Decomposition via the include relationship* (FD-include) [6, 7]: a use case is decomposed into several inclusion use cases that have no direct association with any actor and provide no observable result of their own. Detecting such instances requires understanding both the syntactic structure of the diagram and the semantic intent of the include relationship, a task that benefits from natural language reasoning as well as structural analysis.

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Existing approaches to antipattern detection in UML rely primarily on model-checking rules, OCL constraints, or pattern-matching heuristics [4, 8, 9]. These methods are brittle when diagrams do not conform to an expected representation, and they cannot generalize to novel antipattern types without expert-authored rules. Recent advances in large language models (LLMs) offer a promising alternative. Models trained on sufficiently representative examples can generalize structural reasoning to unseen diagrams.

However, assembling labeled training data for UML antipatterns is expensive and scarce in practice. Published datasets of annotated use case diagrams are small and cover few domains. We address this bottleneck by using Claude Opus, a state-of-the-art LLM, as a controlled diagram author to produce a large, diverse, and precisely labeled set of diagram pairs. This data is then used to fine-tune Qwen2.5-Coder-3B-Instruct, a compact open-source LLM suitable for deployment in resource-constrained settings.

This paper presents a prompt-engineering framework that instructs an LLM to generate syntactically valid, structurally diverse, and correctly labeled pairs of antipattern and refactored PlantUML diagrams. Using this framework, we produced a dataset of 388 annotated samples spanning 194 application domains, each diagram containing 1–5 antipattern instances. Building on this data, we fine-tune a 3B-parameter LLM that achieves 91.0% detection accuracy on held-out test samples.

The remainder of this paper is structured as follows. Section 2 reviews relevant background. Section 3 describes the data generation pipeline. Section 4 characterizes the resulting dataset. Section 5 details the fine-tuning configuration. Section 6 presents evaluation results. Section 7 applies the model to a real-world case study. Section 8 discusses findings and limitations. Section 9 surveys related work. Section 10 concludes.

2 Background

2.1 UML Use Case Diagrams and Antipatterns

A UML use case diagram represents the functional requirements of a system as a set of use cases (ellipses) associated with one or more actors. Three relationship types express dependencies between use cases: the «include» relationship indicates that the base use case always incorporates the behavior of the inclusion use case; «extend» represents optional behavior that may be inserted at an extension point; and generalisation captures inheritance between actors or between use cases, indicating that one element is a specialized form of another.

Antipatterns in use case models are recurring modeling mistakes that violate the intended semantics of these elements [7]. This work focuses on FD-include, where a use case is divided into several inclusion use cases that are not directly associated with any actor. Offering no observable result to a user, they represent internal sub-steps rather than independent services. The correct refactoring merges these inclusions back into the base use case. Concretely, an inclusion use case participates in an FD-include instance if and only if: (i) it is included by exactly one base use case; (ii) it has no direct actor association; (iii) it neither includes nor extends any other use case; (iv) it is not extended by any other use case; and (v) it is not involved in any generalisation.

2.2 LLMs for Diagram Generation and Fine-Tuning

LLMs have demonstrated strong capabilities in generating structured text conforming to domain-specific syntaxes, including source code, configuration files, and markup languages. PlantUML is a text-based diagram description language; its grammar is sufficiently regular that frontier models can produce syntactically valid diagrams with little additional guidance. This has been demonstrated in security modeling contexts, where ChatGPT-5 has been shown to generate misuse case diagrams directly in PlantUML from textual security requirements [2], and ChatGPT has been evaluated for generating STRIDE data flow diagrams, with findings covering syntactic correctness, semantic accuracy, and trust-boundary placement [1].

Adapting a pre-trained LLM to a specialised task such as antipattern detection, without retraining all weights, is achieved through parameter-efficient fine-tuning (PEFT). Low-Rank Adaptation (LoRA) [10] is a PEFT technique that injects trainable low-rank matrices into each weight matrix of a frozen pre-trained model. With rank $r \ll d$, where d is the dimension of the weight matrix, LoRA reduces the number of trainable parameters by orders of magnitude while preserving performance comparable to full fine-tuning.

3 Data Generation Pipeline

3.1 Pipeline Overview

The pipeline takes two inputs. The first is an antipattern specification that names the antipatterns to be instantiated, providing a natural-language description and a refactoring strategy for each. The second is a pool of 194 application domains spanning everyday (e.g. Banking System, Online Store), moderately common (e.g. Tutoring Marketplace, Crowdfunding Platform), and niche domains (e.g. Irrigation Control System, Funeral Service Management System), ensuring the model sees a wide range of domain vocabularies. These inputs drive three pipeline stages: LLM-driven diagram generation, quality review, and training sample packaging, as illustrated in Figure 1.

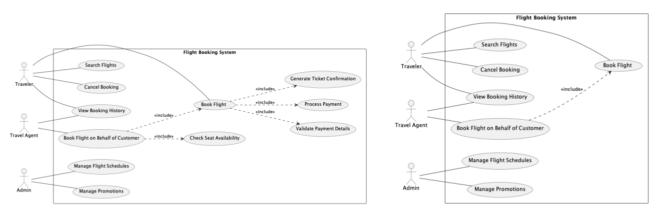
Antipattern configuration + Domain configuration
↓
Diagram generation (Claude Opus API)
↓
2×(PlantUML + detection label) per domain
↓
Quality review
↓
Training set

Figure 1: High-level data generation pipeline.

3.2 Prompt Design

Each diagram is generated using a two-part prompt: a *system prompt* that governs the entire session and remains constant across all diagrams, and a *user prompt* issued once per domain. The system prompt defines how FD-include instances are identified and counted, what counts as a UML element, the diagram validity rules, the refactoring scope, and the required response format.

FD-include instance definition. The system prompt instructs Claude to identify FD-include instances using the five criteria defined in Section 2. Each qualifying (base, inclusion) pair constitutes one instance; a base use case with N qualifying inclusions contributes N separate instances. Chain inclusions (A includes B, B includes C) produce two instances, (A, B) and (B, C), since B is simultaneously an inclusion use case for A and a base use case for C. All instances that share the same base use case form an *instance group*; a diagram in which k base use cases are each decomposed into qualifying inclusions contains k instance groups. Figure 2 shows a Flight Booking System example with two instance groups: *Book Flight* decomposes into three inclusion use cases (*Validate Payment Details*, *Process Payment*, *Generate Ticket Confirmation*), and *Book Flight on Behalf of Customer* decomposes into one (*Check Seat Availability*). All four are FD-include instances and are dropped in the refactored version. The remaining «include» from *Book Flight on Behalf of Customer* to *Book Flight* is retained because *Book Flight* is directly associated with an actor and provides an observable result.



(a) Antipattern: two FD-include instance groups.

(b) Refactored: all four FD-include instances removed.

Figure 2: Flight Booking System: antipattern and refactored diagrams.

Element definition. The system prompt defines a UML *element* as any of four types: actor, use case, «include» arrow, or «extend» arrow. Diagram size is expressed as a total element count, with the target for each domain drawn from the size tier range. The two tiers in Table 1 are deliberately non-contiguous. Small diagrams contain at most one instance group, while medium diagrams are large enough to support up to three. The gap between the tiers ensures structural distinction is preserved regardless of minor count deviations, since Claude may add or omit a few elements to produce a coherent domain model. The ranges therefore serve as target guidelines rather than strict constraints.

Structural validity rules. Several rules enforce UML correctness. Every diagram must include a system boundary rectangle enclosing

Table 1: Diagram size tiers used during generation.

Size	Element range	Max instance groups
small	9–13	1
medium	24–30	3

all use cases, with all actors placed outside it. Every use case in both versions must be associated with at least one actor; a use case with no actor association is invalid. Actor-to-use-case associations must be undirected lines, and «extend» arrows must point from the extension use case to the base use case.

Refactoring scope. The system prompt encodes the refactoring strategy supplied in the antipattern specification. The refactored version must be derived strictly from the antipattern version, only elements that participate in FD-include instances may be removed, and no new use cases, actors, or associations may be introduced. This constraint ensures that the two versions are comparable and that the refactored version contains no spurious domain enrichment.

Response format. The system prompt instructs Claude to produce two PlantUML diagram versions: one with FD-include instances embedded and one with those instances refactored away. It also requires an *antipattern detection label*, a JSON-formatted structured output recording whether the antipattern is present, which elements are involved in each instance, and a natural-language explanation. The schema is illustrated in Listing 1; refactored versions carry a fixed negative label (`"detected": false`, empty `antipatterns` list). Each element is identified by its display name followed by its PlantUML identifier in parentheses (e.g. `UC1`), enabling a downstream modeling tool to locate and refactor the specific use case without ambiguity.

```
{
  "detected": true,
  "antipatterns": [{
    "antipattern_name":
      "Functional Decomposition: Using the
      include relationship",
    "instance_count": 1,
    "instances": [{
      "elements": [
        "Withdraw Cash (UC1)",
        "Validate Account (UC2)"
      ],
      "explanation": "'Validate Account' (UC2)
        is included by 'Withdraw Cash' (UC1),
        has no direct actor association
        ... making it a functional decomposition."
    }]
  }]
}
```

Listing 1: Example antipattern detection label for a Banking System.

The user prompt is issued once per domain and specifies the target domain and a target element count drawn from the size tier range. The Claude Opus 4.6 API is called with this prompt; the response yields a positive sample (the antipattern PlantUML source with its detection label) and a negative sample (the refactored PlantUML source), which are then submitted for manual review. Once approved, each sample is serialized as a conversational training record containing three messages: a fixed system prompt, a user message carrying the PlantUML source, and an assistant message containing the antipattern detection label. Listing 2 shows a condensed positive sample; the corresponding negative (refactored) sample is identical in structure but with `"detected": false` and an empty `"antipatterns"` list.

```
{
  "messages": [
    {
      "role": "system",
      "content": "You are an expert in UML use case
        diagram analysis ..."
    },
    {
      "role": "user",
      "content": "Analyze the following PlantUML use
        case model and detect antipatterns. ...
        @startuml
        ...
        @enduml"
    },
    {
      "role": "assistant",
      "content": "{ detected: true, ... }"
    }
  ]
}
```

Listing 2: Condensed positive training sample.

3.3 Quality Review

Each generated pair is manually reviewed. Despite the detailed prompt constraints, Claude does not always produce fully conformant diagrams; observed issues include use cases with no actor association, violation of the system boundary rule, refactored versions that retain FD-include elements or introduce new use cases absent from the antipattern version, and extension arrows pointing in the wrong direction. When issues are found, the reviewer notes them as free-text descriptions of the specific violations. Those notes are then passed to Claude as part of the user prompt, asking it to regenerate the pair while correcting the identified issues. This loop typically resolves violations within one or two additional attempts.

4 Dataset Characteristics

The pipeline produced 194 antipattern/refactored pairs spanning as many distinct application domains (85 small, 109 medium), yielding 388 training samples in total. One pair per domain ensures the model generalizes across domain vocabulary rather than memorising domain-specific patterns. Table 2 summarises the structural properties of the generated diagrams.

The difference in mean element count between antipattern (15.3) and refactored (11.4) versions directly reflects the refactoring: removing FD-include instances eliminates inclusion use cases and their «include» arrows, reducing total elements by approximately 4 per diagram on average. Only 16 of 194 refactored diagrams retain any «include» relationships, confirming that the generation pipeline correctly preserves only legitimate includes (those whose

Table 2: Structural statistics of the generated diagram dataset.

Metric	Antipattern	Refactored
Samples	194	194
Total elements (mean)	15.3	11.4
Use cases (mean)	10.2	8.2
Actors (mean)	2.9	2.9
Include relationships (mean)	2.1	0.1
Instances per sample (mean)	1.96	—
Instances per sample (range)	1–5	—
Diagrams with size *small*	85	85
Diagrams with size *medium*	109	109

inclusion target is associated with an actor or is included by multiple use cases). The antipattern instance distribution (1–5 instances per diagram) provides the model with varied label complexity, including cases where multiple base use cases each decompose into multiple sub-steps.

5 Fine-Tuning

We apply PEFT to Qwen2.5-Coder-3B-Instruct [11], a 3B-parameter LLM from the Qwen2.5-Coder series. Its pretraining corpus includes large quantities of code and structured text, making it well-suited to PlantUML syntax and structured output formatting. Low-Rank Adaptation (LoRA) [10] adapters are applied to all attention and feed-forward projection matrices with rank $r = 8$ and scaling factor $\alpha = 16$, training 0.48% of total parameters. Samples are formatted using the Qwen2.5 chat template with system, user, and assistant turns, with a maximum sequence length of 2048 tokens. The dataset is split into training and test sets using a domain-stratified 80/20 partition, where entire domains (both antipattern and refactored samples) are assigned to either split, ensuring no domain appears in both. This yields 155 training domains (310 samples) and 39 test domains (78 samples). The model is trained for 6 epochs; training loss converged smoothly from above 1.0 to a final value of 0.18, as shown in Figure 3.

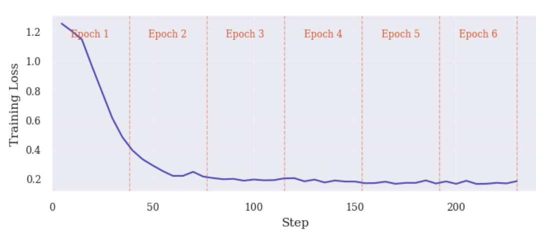


Figure 3: Training loss during LoRA fine-tuning on the generated dataset.

6 Evaluation

Test samples are evaluated for detection accuracy and instance classification. Detection accuracy is the fraction of diagrams for which the model correctly predicts the `detected` field of the detection label; for positive predictions, the antipattern name is also verified. The model correctly predicted the presence or absence of an antipattern in 71 of 78 test diagrams, yielding a detection accuracy of 91.0%, with the correct antipattern name predicted in all positive cases.

Instance classification compares the predicted instance list against the detection label and assigns TP, TN, FP, and FN counts as follows. A negative sample (refactored diagram) predicted as negative contributes one TN; any instance the model predicts on a negative sample contributes one FP. For a positive sample, each expected instance that the model correctly predicts contributes one TP, each missed instance contributes one FN, and each predicted instance with no match in the expected list contributes one FP. For example, if a positive sample expects two instances and the model predicts two but only one matches, the result is 1 TP, 1 FP, and 1 FN. Matching checks that `elements` exactly matches the expected list and that the `explanation` is semantically correct; all results were manually verified to resolve ambiguous cases. Precision, recall, F_1, instance classification accuracy, and Matthews Correlation Coefficient (MCC) are derived from these counts. Table 3 reports the results over 126 verified classification outcomes.

Table 3: Instance-level classification results over 126 outcomes across 78 test samples.

	Pred. +	Pred. −
Actual +	68 (TP)	10 (FN)
Actual −	16 (FP)	32 (TN)

Metric	Value
Precision	0.81
Recall	0.87
F_1	0.84
Accuracy	0.79
MCC	0.55

In antipattern detection, recall is the more critical metric, since a missed antipattern remains silently in the model whereas a false positive can be dismissed by the modeler on inspection. The model achieves a recall of 0.87, recovering the majority of expected FD-include instances, which is the desirable outcome in this context. Precision at 0.81 reflects some false-positive tendency, as the model occasionally flags an FD-include instance where none was intended, but this is an acceptable trade-off. The MCC of 0.55 confirms a moderate-to-strong positive correlation between predictions and ground truth, despite the modest dataset size.

Manual inspection of the 16 FP and 10 FN instances reveals three recurring FP patterns and two FN patterns. Roughly half of all FPs involve prediction of an FD-include instance whose inclusion-side use case is directly associated with an actor in the diagram, violating the criterion that disqualifies it from FD-include membership. This error is especially consistent when the overlooked actor is a system entity rather than a human role. In two unrelated booking domains a payment use case is simultaneously included by a base use case and directly associated with a Payment Gateway actor, yet the model flags the relationship as FD-include in both cases. This actor-association error persists across both the antipattern and refactored versions of the same domain in three cases, confirming a systematic gap in the model's actor-association check rather than a stochastic failure. A second FP pattern is extend-for-include confusion, where an «extend» relationship is misread as an FD-include instance. A third, less frequent pattern involves structurally unrelated use case pairs. Here the model asserts an include link between two use cases that share no relationship in the diagram, suggesting occasional hallucination of structural connections. Among FNs, the majority are straightforward omissions in diagrams where other instances in the same diagram were correctly identified, pointing to incomplete structural scanning rather than misclassification of relationship type. One FN involves a chain FD-include where A includes B and B includes C; the second link (B, C) is missed even though the first (A, B) is detected, likely because the intermediate node B plays a dual role as both an inclusion use case and a base use case.

7 Real-World Case Study: MAPSTEDI

To assess the model's ability to generalise beyond synthetically generated diagrams, we apply it to a subsystem of MAPSTEDI, a distributed biodiversity database system [15] that has been used as a benchmark in prior antipattern detection research [6, 7]. The subsystem modelled here covers database access, database integration, database editing, and administrative processes, comprising 16 use cases and 7 actors.

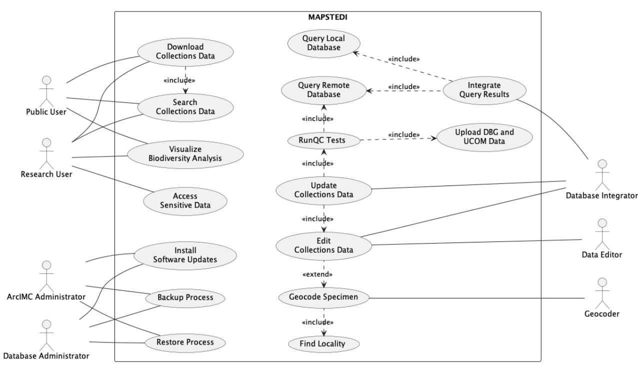


Figure 4: MAPSTEDI subsystem use case diagram used as the real-world evaluation input.

Figure 4 shows the use case diagram. The three ground-truth FD-include instances, established by manual inspection against the antipattern definition, are: (1) *Integrate Query Results* → *Query Local Database*, where *Query Local Database* has no direct actor association and represents an internal query sub-step; (2) *RunQC Tests* → *Upload DBG and UCOM Data*, where *Upload DBG and UCOM Data* has no actor association and is not reachable independently; and (3) *Geocode Specimen* → *Find Locality*, where *Find Locality* similarly has no actor association and is included by only one use case.

The model produced 5 predicted instances, yielding 1 TP, 4 FP, and 2 FN (Precision = 0.20, Recall = 0.33). The single correct prediction was instance (1). The 4 FPs expose three failure modes already observed in the synthetic test set, compounded by the diagram's higher complexity. Two FPs flag use cases that carry direct actor associations (*Search Collections Data* associated with Public and Research Users; *Edit Collections Data* associated with Database Integrator and Data Editor), repeating the actor-association check failure documented in Section 6. One FP flags *Query Remote Database*, which is included by *two* base use cases and therefore represents legitimate shared behaviour rather than functional decomposition. One FP flags *RunQC Tests* despite it having an outgoing include to *Query Remote Database*, violating the criterion that a decomposed use case must not itself include other use cases. The 2 FNs, instances (2) and (3), are straightforward omissions of structurally unambiguous FD-include cases, consistent with the incomplete-scanning FN pattern in the synthetic results.

The degraded performance on MAPSTEDI relative to the synthetic test set is consistent with the synthetic distribution shift discussed in Section 8. The training diagrams are smaller (median 8 use cases), syntactically uniform, and contain at most one antipattern type per diagram, whereas MAPSTEDI is a practitioner-authored diagram with 16 use cases, shared-behaviour patterns, and multi-actor use cases not represented in the training distribution. The case study therefore confirms the need for real-world diagram examples in the training set and motivates the dataset extension directions outlined in Section 8.

8 Discussion

The results validate that a frontier LLM can serve as a reliable diagram labeler, as all generated pairs passed manual review, with violations corrected through targeted regeneration, and the structural statistics confirm that refactored versions faithfully remove antipattern elements without introducing spurious additions (16 out of 194 retain legitimate includes, consistent with the prompt intent). The model achieves competitive detection performance with only 3 billion parameters and < 0.5% of its weights updated, confirming that synthetic data generation combined with parameter-efficient fine-tuning enables effective specialisation of a compact LLM at modest computational cost.

The current dataset targets only FD-include; the full antipattern catalogue [6, 7] covers many further types (e.g. FD-extend, Accessing an Extension Use Case), and generalisation to any of these would require repeating the generation and fine-tuning process, or jointly training on multiple antipatterns simultaneously. A second concern is synthetic distribution shift, as Claude Opus exhibits systematic stylistic preferences that may differ from practitioner-authored diagrams, so performance on real-world inputs remains to be evaluated. Generation is also capped at 5 antipattern instance groups per diagram, whereas real-world diagrams may contain more. The FN pattern of incomplete scanning identified in the error analysis suggests performance may degrade further as instance count grows. The model was trained and evaluated exclusively on PlantUML-formatted diagrams, so its applicability to other use case diagram notations or modeling tools remains untested. Finally, 78 test samples across 39 domains yield somewhat unstable metric estimates, and a larger held-out set would provide more reliable performance bounds.

The most immediate extension is broadening the dataset to cover multiple antipattern types simultaneously, working towards a unified model capable of detecting a full antipattern catalogue. Expanding the diagram size range to include larger diagrams would stress-test the model on more complex inputs. Evaluating on a handcrafted benchmark of real practitioner diagrams would measure how well the model generalizes beyond synthetically generated examples. A natural progression of the task is to move from detection to refactoring, where the model not only identifies antipattern instances but also outputs the corrected diagram.

9 Related Work

A metric-based approach for antipattern detection in UML designs [8] targets class and sequence diagrams. Their approach detects five antipatterns including Functional Decomposition by computing OO quality metrics such as coupling, cohesion, and complexity, and comparing them against predefined thresholds. While it operates at the UML design level, the approach requires manually defined thresholds. Our approach differs by learning detection from examples rather than from metrics or rules, and targets use case diagrams directly from their PlantUML source.

The most comprehensive catalogue of use case model antipatterns to date [7] identifies 26 recurring modeling defects (including FD-include targeted in this work) and codifying them as machine-checkable heuristics within ARBIUM, a supporting tool. Detection in ARBIUM is semi-automated via OCL-style queries; however, refactoring suggestions are made textually and applied by the developer manually. Our work draws on their antipattern taxonomy and takes the next step by replacing the rule-based detector with the model, removing the need for hand-crafted detection rules and enabling detection from plain-text PlantUML sources without requiring the diagram to be serialised in XMI.

Subsequent work [13] extends the ARBIUM line by automating the refactoring step through model transformations: OCL queries identify antipattern instances, and ATL transformations then apply the prescribed structural repairs. While this fully automates the detect-and-refactor loop for a subset of the catalogued antipatterns, the approach still relies on expert-authored formal rules and requires diagrams to be serialised in XMI rather than plain-text sources. Our approach is complementary: by training an LLM on labeled diagram pairs, we show that the detection task can be learned from examples alone, without hand-crafted rules.

Early assessments of ChatGPT's ability to generate UML class diagrams found syntactic and semantic limitations in models of that era [5]. More recent studies show frontier models performing markedly better on structured diagram generation tasks [1, 2]. Our work builds on this capability but uses it for labeled pair generation rather than description-to-diagram translation.

10 Conclusion

We have presented an end-to-end pipeline for fine-tuning an LLM to detect the Functional Decomposition antipattern in PlantUML use case diagrams. The pipeline uses Claude Opus as a synthetic diagram author to generate 388 high-quality labeled training samples across 194 application domains, validated through a structured manual review workflow. Qwen2.5-Coder-3B-Instruct, fine-tuned with LoRA on this dataset, achieves 91.0% detection accuracy on a domain-held-out test set, demonstrating that PEFT on LLM-generated synthetic data is an effective strategy for specializing compact models on structured UML reasoning tasks. The pipeline is designed to scale, as additional antipattern types can be added to the configuration and the generation process repeated, extending the detector to new antipattern types without any manual data collection. The dataset, generation scripts, fine-tuning notebook, and evaluation results are publicly available [12].

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